

CSC 4553

Fall 2026 - Day 2

Admin

- Lab 1 due Friday, 18:00
- Quiz 1 due Monday, 18:00

OS abstractions

Questions to consider

- What are the main abstractions OSes provide?
- What are the abstraction challenges?
- How does the OS manage access to resources?

Abstractions

- Abstractions *simplify application design* by:
 - hiding undesirable properties,
 - adding new capabilities, and
 - organizing information
- Abstractions provide an **interface** to application programmers that separates **policy**—what the interface commits to accomplishing—from **mechanism**—how the interface is implemented.

What are the abstractions?

- CPUs
 - Processes, threads
- Memory
 - Address space
- Disk
 - Files

Example OS abstraction: file systems

- What **undesirable properties** do file systems hide?
 - Disks are slow!
 - Chunks of storage are actually distributed all over the disk
 - Disk storage may fail!
- What **new capabilities** do files add?
 - Growth and shrinking
 - Organization into directories, searchability
- What **information** do files help organize?
 - Ownership and permissions
 - Access time, modification time, type, etc.

Abstraction tradeoffs - discussion

- Identify undesirable properties hidden by, new capabilities added, and info organization provided with these abstractions:
 - Process / threads
 - Address space

Abstraction pros / cons

- **Advantages** of OS providing abstractions?
 - Allow applications to reuse common facilities
 - Make different devices look the same
 - Provide higher-level or more useful functionality
- **Challenges?**
 - What are the correct abstractions?
 - How much should be exposed?

OS design requirements - what do we need?

- **Convenience**, abstraction of hardware resources for user programs
- **Efficiency** of usage of CPU, memory, etc.
- **Isolation** between multiple processes
- **Fairness** of access to contended resources
- **Reliability**, the OS must not fail
- Other:
 - **Security**
 - **Mobility**

Another view of the same list: the OS as a **resource manager** that shares resources “well.”

Challenges? What are the correct **mechanisms**? What are the correct **policies**?

What isn't clear?

Comments? Thoughts?